Schrödinger’s Cat Explained [B2]

Il famoso esperimento del “gatto di Schrödinger” è stato ideato dal fisico Erwin Schrödinger nel 1935 per dimostrare che, se le leggi del mondo subatomico venissero applicate letteralmente al mondo macroscopico, arriveremmo a conclusioni ridicole.

Quantum physics is famously incomprehensible and perplexing for **laypeople** because it often contradicts the natural world we perceive with our senses. To highlight what he saw as a fundamental absurdity in this field, Austrian physicist Erwin Schrödinger (1897-1961) **devised** a thought experiment in 1935 that came to be known as “Schrödinger’s cat”. In it, a cat is placed in a sealed box with a vial of poison, and a **device** containing a tiny amount of radioactive material and a Geiger counter (which detects radiation.) If a single atom in the sample **decays**, the counter **triggers** a mechanism — often described as a hammer — that breaks the vial, **releasing** the poison and killing the cat. According to quantum mechanics, until an observer opens the box to measure the system (meaning to check on the cat), the radioactive atom exists in a superposition of having both decayed and not decayed. Consequently, Schrödinger argued that by the logic of the theory, the cat itself would be forced into a superposition state: existing simultaneously as both alive and dead. Culture

Bizarre Science: Ig Nobel Prize È arrivato il momento della cerimonia di consegna degli Ig Nobel, i premi assegnati ai dieci più insoliti, bizzarri e irrilevanti risultati scientifici di quest’anno! Talitha Linehan , Molly Malcolm

REDUCTION TO ABSURDITY

Schrödinger proposed this scenario as a reductio ad absurdum to illustrate that quantum rules for subatomic particles lead to ridiculous conclusions when applied to large, macroscopic objects. In quantum theory, before a system is measured, it can only be described by a wave function, a mathematical entity that **encodes** the probabilities of different **outcomes** rather than a single, definite reality. If you want to know more about this topic, read the article Quantum Technology: The World Of Very Small Things.

Glossary

- **laypeople** = profani, non addetti ai lavori
- **devised** = ideare
- **device** = dispositivo
- **decays** = disintegrarsi
- **triggers** = innescare
- **releasing** = rilasciare
- **encodes** = codificare
- **outcomes** = risultati